

DASE7140 Final Project

Object Tracking

1 Description

Object tracking is a common computer vision task. It tracks moving objects over time in a video or a sequence of images and assigns unique IDs to them so they can be followed across frames. This final project about object tracking can be either “easy” or “difficult”: the video clip `sample.mp4` contains many pedestrians, and the task is writing a program to track **the specific person** highlighted in the bounding box below for this video—only this person, but no others.



The “easy” part is that you are allowed to use any AI tools or open-source code to finish the project. If you are familiar with this area, it will take you only several minutes. Yes, it’s easy to get a pass for this course.

The “difficult” part is that, as you may already know, we have a limited number of Grade-A scores, so this final project is intentionally designed as an open request, and we will rank the submissions according to their completeness, performance, and technical highlights. Just consider that this is a requirement from customers without machine learning background, usually they don’t exactly know what they want, but if your deliverables do better than their expectations, your effort will be much appreciated. My original thought was to leave some hints here, but my second thought was, it’d be a good practice to put oneself in the customer’s position—What would you want for an

object-tracking tool as a customer? Sometimes, you may even come up with great ideas that the customers have never imagined, and you lead the trend.

This final project is weighted 50 points (the previous three assignments are 10, 20, and 20 points, respectively). The students' final scores will be normalized to account for the Grade-A (≥ 75 points) quota. **Please invest your time wisely** (I don't recommend putting too much or too little effort into this project)—as in our lives, there are always far and near goals, and we may need to take approximately optimized actions.

2 Required deliverables

You are required to submit four files:

1. **One Python environmental description file:** `requirements.txt`. Please use `pip freeze` to generate this file¹.
2. **One model file** used for object tracking. You can freely choose any format and filename for the model file, however, **the file size should be less than 10 MB**.
3. **One processing Python script:** `track.py`. This script, the model file, and `sample.mp4` will be put in the same folder. The TAs will run `python track.py` in that folder under a virtual environment². After running `track.py`, **a result video `result.mp4` (or with any other possible video file extensions like `.avi`, `.mkv`, etc.) should be generated in the same folder**. The result video should show the bounding box or the highlighted segmentation mask for the specific person, so the TAs can evaluate the tracking result. To select the specific person, you may hardcode the relevant information in `track.py`.
4. **One PDF file `highlights.pdf`** to list the technical highlights of the deliverables. Please be concise and to the point.

¹I know it's old-fashioned, but let's keep it this way.

²No worries. The TAs will try to solve some minor "missing packages" problems.